(ii) the signal attenuation per km for the fiber.

(iii) the overall signal attenuation for a 10 km optical link using the same fiber with splices at 1 km intervals each giving an attenuation of 1 dB.

(iv) the numerical i/p / o/p power ratio in (C).

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7

(d) What do you mean by attenuation? Discuss all types of attenuation that occur in optical fiber communication. 7

Unit-V

5. (a) Define population inversion. Also specify application. 2

(b) Compare the electrical and optical Bandwidth for an optical fiber communication system and develop a relationship between them. 7

(c) With the aid of suitable diagram explain the principle of operation of Injection Laser. 7

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(d) Differentiate between spontaneous emission and stimulated emission as applied to LASER diode. Compare the performance of LED and Laser Diode. 7

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B. E. (Eighth Semester) Examination, April-May, 2016

(New Scheme)

(Et & T Engg. Branch)

ADVANCED COMMUNICATION SYSTEMS

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Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. Part (a) of each unit is compulsory carrying 2 marks. Attempt any two parts from (b), (c) and (d) of each unit with each carrying 7 marks.

Unit-I

1. (a) Differentiate between sub synchronous and super synchronous satellite. 2

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(b) What is meant by look angles? Explain them with reference to GEO satellite and the earth station. Derive the expression for look angles. 7

(c) Determine slant range and visibility angle. Show that atleast 3 GEO's are required to cover entire earth surface. 7

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(d) List various frequency bands being used in satellite communication. Compare the advantages and disadvantages of different band considering the effect of propagation media. 7

Unit-II

2. (a) Define Noise figure. What is the relation between noise figure and noise temperature? 2

(b) Why is satellite link design required? Also derive the general link design equation for communication satellite. 7

(c) What are various interferences that may affect the link performance of a satellite? 7

(d) What is Iriss Transmission equation? Derive it including all the losses. 7

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Unit-III

3. (a) What is guard time? 2

(b) Compare the performance of FM/FDM, SCPC and CSSB system. 7

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(c) With the help of a structure define the TDMA super frame and explain how it differs from a simple TDMA frame. 7

(d) What is burst? Differentiate between the Reference burst and Traffic burst with diagram. 7

Unit-IV

4. (a) Define Meridional rays and skew rays. 2

(b) Explain with the aid of diagram :

(i) the multimode step index fiber

(ii) the single mode step index fiber

Compare the advantage and disadvantage of these two type of fiber for use as a optical channel.

(c) When the mean optical power launched into an 8 km length of fiber is 120 μw, the mean optical power at the fiber optical power is 3 μw.

Determine :

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(i) the overall signal attenuation or loss in dBs through the fiber assuming there are no connector or splices.